

GENERAL ECOLOGY

BIOLOGY 541

Fall 2025

Instructor: Dr. Sue Cooke
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Office/Phone: P569, 603-641-4328
Office Hours: By appointment
Class meets: M9:10-12, P505

Lab Instructor: Sydney Rollins
Email: sydney.rollins@unh.edu
Lab meets: Wednesday 4-7pm

Objectives:

Upon completion of this course, you should have acquired a basic understanding of evolution, environmental factors, communities, populations, ecosystems, and life history traits.

Prerequisites:

BIOL 413-414 or equivalent

Textbook:

R. Relyea. 2021. *The Economy of Nature*, 9th ed. W. H. Freeman and Co, ISBN:9781319245542

Additional materials: Hofmann, A, H. 2021. *Writing in the Biological Sciences*, 4th edition. Oxford University Press. ISBN 9780197543580.

Course Modality and Designation:

This class will be held face to face each week on Monday from 9:10-12. Materials and assignment links for the course will be posted in weekly modules in MyCourses. Modules will be available Monday at 12:01am each week. *This is a writing intensive course and satisfies the WI requirement within the biology major.*

Credit Hours & Time Commitment:

This is a 4-credit course that involves both lecture and lab activities/assignments. According to the UNH [Credit Hour Policy](#), a 4 credit course equals 12 hours of student academic work per week for a 15 week course. You will spend 3 hours per week in lecture and 3 hours per week in lab. You should therefore plan to spend approximately 6 hours per week outside of class completing assignments and reviewing materials to prepare for assessments.

Classroom Policies

Evaluation:

- 3 Lecture exams (each worth 20% of semester grade)
- Lab assignments (30% of semester grade)
- Paper presentation and discussions (10% of semester grade)

You will receive one combined grade for your work in lecture and lab for this course. As you can see outlined above, the lab makes up 30% of the final semester grade. However, to pass this course, ***you must pass both the lecture portion and the lab portion of the course independently***. If your performance in either portion of the course is below the required grade to pass the course for the major (C-, or 70%), you would not be able to receive credit for the course.

Lab assignments will be discussed in your first lab period. For the lecture, your points come from exams and paper set discussions. You will take 3 non-cumulative exams over the course of the semester (see course schedule). All

exams will be taken in class, and all exams must be completed during the first half of the class period (1.5 hours). Exams will be followed by a break and then lecture.

Four times during the semester we will discuss some classic articles from the primary literature in ecology. Each student will be responsible for presenting a summary of one or more of the papers and leading a discussion. Each student will receive a grade based on the quality of their presentation and their participation in the discussions throughout the semester.

The grade scale is: 94-100%=A; 90-93%=A-; 87-89%=B+; 84-86%=B; 80-83%=B-; 76-79%=C+; 73-75%=C; 70-72%=C-; 66-69%=D+; 63-65%=D; 60-62%=D; Below 59.99%=F.

Attendance:

In general, attendance is mandatory because it is strongly correlated with success in this course! This is a fast-paced class, and we will cover a significant amount of content during each meeting. I will take attendance at the start of each class. If you arrive late, you are responsible for checking in with me before leaving class. This is your responsibility. I will not accept emails after class as proof of attendance! You should make every effort to be on time for class because walking in late is disturbing to other students. We understand there may be unavoidable circumstances that prevent you from attending. If you cannot attend a lecture or a lab for any reason, please let us know as far in advance as possible and we will attempt to accommodate you.

Classroom behavior

You are expected to always conduct yourself in a professional manner while participating in this course. This includes an expectation of respectful behavior during any online or face to face interactions with classmates or me. Regarding cell phones and other electronic devices, please turn off your cell phones in class! If you MUST have your cell phone on because you are expecting an emergency call or need to be available for personal matters, please keep your cell phone on vibrate and leave the room to answer it. If possible, inform me in advance if you are expecting a call and sit on the end of a row so you minimize disturbances to your fellow students! DO NOT TEXT OR ACCESS FACEBOOK OR OTHER SOCIAL MEDIA DURING CLASS.

Important Academic Policies

Academic Alerts

The University is invested in your academic success. If a faculty member is concerned about your academic behavior or performance, they may submit an academic alert. Academic alerts are not punitive. The goal is to provide you with support and resources to support your success. They act as an important check-in point and, if you receive an academic alert, you will receive an email to your UNH email address. It is strongly recommended that you meet with a professional advisor and connect with your instructor to discuss the reason for the alert.

Accommodations for Disability

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Student Accessibility Services (SAS) office. The Student Accessibility Coordinator at UNHM is Alex Wedge. Please reach out to the SAS office via email Alex.Wedge@unh.edu or registration information and disability related questions.

Academic Integrity policy

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic integrity, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<https://catalog.unh.edu/srrr/student-policies-regulations/academic-integrity/>) By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as submitting someone else's work – in whole or in part – as your own; collaborating on answers for individual assignments or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an in-person exam or referring to notes, the textbook, or any prohibited device such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of an in-person exam. Ear buds are not permitted.

For online exams, you will be given instructions about which materials are and are not permitted to be used. You will be asked to digitally sign an Academic Honesty statement before beginning any exam agreeing to follow all exam policies.

Posting any portion of faculty-created materials such as homework assignments or exam questions to "tutoring" or "study" websites is a serious violation of the Academic Honesty policy of the University. Any instances of this form of cheating will be brought directly to the department chair and dean's office.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Referral Resources for Students

Library

The UNH Manchester librarians are available to assist you with the research process. Visit the library's website at <https://cps.unh.edu/library> to learn about library services and to search for reliable academic sources. You can contact the library at 603-641-4173 or at unhm.library@unh.edu.

The links below guide you to useful online library resources:

- Make a **Research Appointment** with a librarian: <https://libraryguides.unh.edu/remotearchive/researchhelp>
- Use the **Library Search Box** to find information: https://libraryguides.unh.edu/librarysearchbox_unhmanchester
- Reserve a **Study Room**: <https://cps.unh.edu/library/support-services>
- Discover resources for **Citing Sources**: <https://libraryguides.unh.edu/unhmcitingsources>
- Learn strategies for **Evaluating Sources**: <https://libraryguides.unh.edu/ENGL401UNHManchester/evaluatingresources>.

Knack Peer Tutoring

Knack is a peer-to-peer tutoring platform available to all enrolled students for all undergraduate courses in Durham and Manchester and is offered at no charge to students. Students looking for additional assistance outside of the classroom are advised to consider working with a peer tutor through Knack. UNH has partnered with Knack to provide students with access to verified tutors who have successfully completed your course. To view available tutors, visit: unh.joinknack.com. Sign in with your student account.

Questions about Knack Tutoring can be sent to Stephanie Kirylych, Director of Advising, at stephanie.kirylych@unh.edu.

Connors Writing Center (CWC)

The [Connors Writing Center](#) offers one-on-one writing conferences to current UNH students (undergraduate and graduate). They work with writers from all disciplines on many kinds of writing. Our conferences are conducted by trained writing assistants who are UNH undergraduate and graduate students. The CWC offers both in-person and virtual appointments, available to any current UNH student. The online sessions are synchronous (real-time) appointments designed to help students have one-on-one conversations about their writing at any stage of the process, mirroring what we do during in-person appointments. [Students can schedule an appointment any time](#)

during the semester.

Online Writing Lab (OWL)

The [Online Writing Lab \(OWL\)](#) assists students via asynchronous remote collaboration with qualified writing assistants. The OWL is not an editing or manuscript preparation service. Feedback may include reference to mechanics on the submitted paper, but it will be targeted to bring attention to patterns and issues to help the writer self-correct the paper and carry forward lessons to improve future writing. Students in all fields of study can submit up to 20 double-spaced pages of any form of writing, including course papers, resumes, creative essays, or personal statements. Within 3 business days, writing assistants provide feedback that can be printed or viewed on screen. The OWL is accessible through any computer that has an Internet connection and Microsoft Word.

To submit a paper, students can sign in at unh.mywconline.com and select the "Online Writing Lab" schedule.

Student Wellness Support- Mental Health Services

In partnership with The Mental Health Center of Greater Manchester, UNH Manchester offers consultation visits on campus and virtually during the academic year with a mental health counselor Monday-Thursday. Students can make counseling appointments directly through an online scheduling system located on the [UNH Manchester Wellness page](#).

Services include:

- Free confidential screening & consultation with a licensed mental health therapist.
- Referrals to mental health or substance misuse treatment.
- And assistance in understanding how to afford additional treatment (with or without insurance!) or find free services.

Basic Needs Resources:

UNH has many resources that can help students who are experiencing food insecurity, housing or other financial concerns. All enrolled students are eligible for Basic Needs resources and services. A complete listing of university and local basic needs resources can be found on the [UNH Manchester Basic Needs webpage](#). The following UNH resources students are referred to most frequently are listed below:

- **The Student Emergency Financial Assistance Fund (SEFA)** provides UNH undergraduate and graduate students unable to meet essential expenses due to a temporary or unexpected hardship with short-term financial assistance once per academic year. *Please note that funding is currently limited due to high demand and UNH had to **pause funding for graduate students** at this time.* We encourage graduate students to schedule a time to meet with the Basic Needs team to discuss their options and connect them to additional resources at basic.needs@unh.edu. UNH received a grant from Granite Edvance to support our Student Emergency Financial Assistance program specifically for undergraduate students. To request SEFA students must complete the [Student Emergency Financial Assistance Request Form](#)
- **UNH Manchester Wildcat Cupboard**, located on the second floor, Room 290, is available to all UNH Manchester undergraduate and graduate students, staff or faculty who need access to free, nourishing food. There are an assortment of items including pasta, canned food, cereal, milk, eggs, fresh fruits and vegetables, which are purchased from the NH Food Bank through a grant or donated by members of the community. The Cupboard is open Tuesdays and Thursdays from 12:30-3:30 pm and Wednesdays from 3:00-6:00 pm. Walk ins are welcomed and no appointments are needed.
- **The UNH Manchester Wildcat Wardrobe** provides students with access to new and gently used professional attire for interviews, career fairs, networking events, and the workplace. Students can visit the Wildcat Wardrobe, located on the second floor in Room 288 (next to the Wildcat Cupboard) Monday-Friday from 9-5 pm. Items are completely free.

Confidentiality and Mandatory Reporting

The University of New Hampshire and its faculty are committed to assuring a safe and productive educational

environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university's Title IX Coordinator (Bo Zarycky, Bo.Zarycky@unh.edu, 603-862-2930/1527 TTY) any incidents of sexual violence and harassment shared by students. All employees are subject to this mandatory reporting. If faculty or staff wish report an incident while on campus they can speak with Manchester's Title IX Deputy Intake Coordinator and Assistant Dean Lisa Enright 603-641-4336. Lisa's office is located on the second floor in the Library and Engagement Center, Room 236. If a student wishes to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, here is a list of confidential service providers. Including a statement of confidentiality and mandatory reporting along with links of supports for students is important for helping our community stay in compliance with Title IX, as well as living up to our commitment of building a safe community for everyone:

- [The SHARPP Center for Interpersonal Violence Awareness, Prevention, and Advocacy](#) at (603) 862-7233/TTY (800) 735-2964. SHARPP's Expanded Services Coordinator, Caroline Young, is available to meet virtually year-round Mondays-Fridays 9am-4pm and is on Manchester's campus Mondays during the Fall and Spring semesters (in Room 471). If you'd like to book a confidential appointment with her [schedule an appointment here](#). SHARPP Advocates are also available 24/7 via our Crisis Line: 603-862-SAFE (7233).
- [REACH Crisis Services](#): 24/7 Support Line: 603-668-2299. SMS/Text (M-F 8:30 AM-5:00 PM): 603-836-9620. For more information about what happens when a student reports, how the university treats your information once a report is made to the Title IX Coordinator, your rights and reporting options at UNH (including anonymous reporting options) [please visit student reporting options](#).
- The [uSafeUS](#) app is also available for students to keep reporting options and resources easily accessible on their phones.

Help us improve our campus and community climate. If you have observed or experienced an incident of bias, discrimination or harassment, please report the incident by contacting the Civil Rights & Equity Office at UNH.civilrights@unh.edu or TEL # (603) 862-2930 voice/ (603) 862-1527 TTY /

Tentative Lecture Schedule:

Date	Topic	Reading
Aug 25 th	Introduction to Course; The Science of Ecology Evolution and Ecology	Chap 1 Chap 7 & Evol. Chapter in Gen. Biol. Text
Sept 1 st	No in-person class...Labor Day! Review The Physical Environment lecture video	Chap 2-5
Sept 8 th	Finish Evolution & Ecology Start Biomes	Chapter 6
Sept 15 th	Finish Biomes Communities, part 1	Chapter 18
Sept 22 nd	Evolution and Natural Selection Intertidal Ecology	Paper Set 1
Sept 29 th	EXAM 1 Communities, part 3	Chap 19, 22-23
Oct 6 th	The Nature of Communities Ecosystems: Energy	Paper Set 2 Chap 20
Oct 13 th	Fall Break	
Oct 20 th	Ecosystems: Nutrient Cycles Population Distributions	Chap 21 Chap 11
Oct 27 th	Trophic Dynamics Lecture catch-up	Paper Set 3
Nov 3 rd	Population Growth & Regulation	Chap 12
Nov 10 th	EXAM 2 Population Dynamics	Chap 13
Nov 17 th	Life History Patterns	Chaps 8-10
Nov 24 th	Species Interactions	Chap 15, 17
Dec 1 st	Competition	Chap 16
Dec 8 th	Competition and the Niche Predation	Paper Set 4 Chap 14
Dec 15 th	EXAM III	

General Ecology Paper Sets

(Papers are available on MyCourses)

Paper Set 1 -- Evolution and Natural Selection

Barash, D. P. 1976. Male response to apparent female adultery in the mountain bluebird: an evolutionary interpretation. *American Naturalist* 110: 1097-1101.

Gould, S. J. and R. C. Lewontin. 1979. The spandrels of San Marco and the panglossian paradigm: A critique of the adaptationist programme. *Proceedings of the Royal Society of London B* 205: 581-598.

Kettlewell, H. B. D. 1955. Selection experiments on industrial melanism in the Lepidoptera. *Heredity* 9: 323-342.

Paper Set 2 -- The Nature of Communities

Clements, F. E. 1936. Nature and structure of climax. *Journal of Ecology* 24: 252-284.

Gleason, H. A. 1926. The individualistic concept of the plant association. *Bulletin of the Torrey Botanical Club* 53: 7-26.

Tansley, A. G. 1935. The use and abuse of vegetational concepts and terms. *Ecology* 16:284-307.

Paper Set 3 -- Ecosystems: Trophic Dynamics

Lindeman, R. L. 1942. The trophic-dynamic aspect of ecology. *Ecology* 23: 399-418.

Cook, R. E. 1977. Raymond Lindeman and the Trophic-dynamic concept in ecology. *Science* 198:22-26.

Teal, J. M. 1962. Energy flow in the salt marsh ecosystem of Georgia. *Ecology* 43: 614-624.

Paper Set 4 -- Competition and the Niche

Grinnell, J. 1917. The Niche-relationships of the California Thrasher. *Auk* 34:427-433.

Hutchinson, G. E. 1957. Concluding remarks. *Population Studies: Animal Ecology and Demography*. Cold Spring Harbor Symposium on Quantitative Biology 22:415-427.

MacArthur, R. H. 1958. Population ecology of some warblers of northeastern coniferous forests. *Ecology* 39:599-619.